
Stock Market Prediction Using Machine Learning

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Abstract

This project investigates the prediction of next-day directional price movements of Apple Inc. (AAPL) using classical supervised machine learning. Using eleven years of daily OHLCV data (January 2015 – May 2026) from Yahoo Finance, eighteen technical indicators were engineered as features — covering moving averages (SMA, EMA), RSI, MACD, Bollinger Bands, return-based signals, and lagged returns — to frame the problem as binary classification: UP (1) if the next day's close exceeds today's, DOWN (0) otherwise. Four models — Support Vector Machine (RBF kernel), Random Forest (200 trees), K-Nearest Neighbours (k=5), and Logistic Regression — were trained under a strict 60/20/20 chronological split with StandardScaler normalisation. The best-performing model, SVM, achieved a test accuracy of 52.1%, marginally above the 50% chance baseline, consistent with the semi-strong Efficient Market Hypothesis. Feature importance analysis identified Return_Lag_3, Open_Close_Diff, Daily_Return, Volume_Change, and RSI as the most discriminative signals.

1. Introduction

Financial markets represent one of the most challenging domains for machine learning, combining non-stationarity, low signal-to-noise ratios, and the self-defeating nature of discovered patterns that are quickly arbitrated away. The Efficient Market Hypothesis (Fama, 1970) posits that historical price information is already reflected in current prices, rendering systematic technical prediction theoretically futile. Yet the availability of deep historical data, advances in ML, and persistent evidence of short-horizon momentum and behavioural biases motivate a rigorous empirical investigation. Apple Inc. (AAPL) — one of the world's most liquid large-cap equities — provides an ideal test bed: a 2,857-day trading history spanning multiple market regimes (the 2017–2019 bull market, the 2020 COVID-19 crash, the 2022 rate-driven correction), balanced class distributions, and well-documented technical dynamics.

The core challenge is framed as a binary classification task with strict temporal integrity: can a model trained exclusively on price-derived technical features learned from the past predict the direction of tomorrow's close? This question carries both scientific significance — testing the limits of market predictability — and practical relevance to algorithmic trading strategy design.

1.1 Objectives

(i) Construct a reproducible ML pipeline for AAPL using publicly available data. (ii) Engineer 18 domain-relevant technical features from raw OHLCV data. (iii) Train and compare four classical classifiers under identical, temporally sound evaluation conditions. (iv) Analyse feature importance to identify the strongest predictive signals. (v) Draw empirically grounded conclusions about market predictability in the context of the EMH.

2. Data Understanding and Preprocessing

Raw OHLCV data was acquired via the yfinance Python library, yielding 2,857 daily observations for AAPL from January 1, 2015, to May 2026. Eighteen features were computed deterministically from Close, High, Low, Open, and Volume columns. Simple Moving Averages (SMA-7, SMA-21) and Exponential Moving Averages (EMA-7, EMA-21) track trend direction. The 14-day RSI quantifies momentum velocity, oscillating between 0 and 100 with overbought/oversold thresholds at 70 and 30 respectively. MACD (difference of 12-day and 26-day EMAs) and its 9-day Signal line measure trend momentum shifts. Bollinger Bands (BB_Upper, BB_Middle, BB_Lower) contextualise price within its 20-day rolling volatility envelope. Return-based features — daily percentage return, volume change, (High–Low)/Close range, and (Close–Open)/Open differential — capture intraday and session dynamics. Four lagged return variables at 1-, 2-, 3-, and 5-day offsets inject temporal autocorrelation structure into the feature matrix.



Figure 1: AAPL Close Price with SMA-7 and SMA-21 (2015–2026)

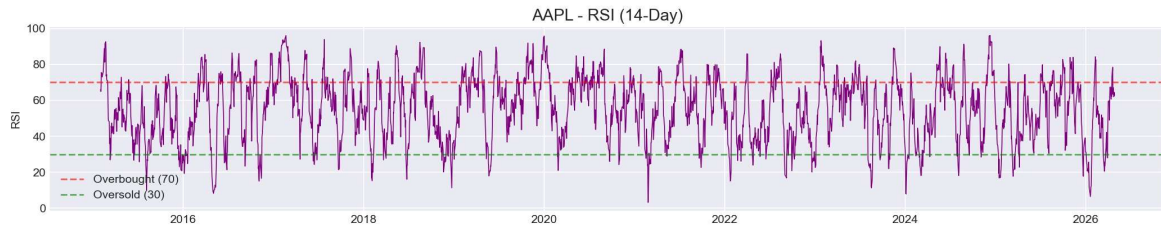


Figure 2: AAPL 14-Day RSI with Overbought (70) and Oversold (30) Thresholds

Following feature computation, all rows containing NaN values (arising from rolling window initialisation and the final shifted target row) were dropped, leaving a clean, fully-populated dataset. The binary target was defined as: Target = 1 (UP) if `next_day_Close > today_Close`, else 0 (DOWN). Class balance was approximately 53% UP / 47% DOWN, mitigating the need for resampling. Data was partitioned chronologically — `shuffle=False` — into 60% training (~1,714 samples), 20% validation (~571), and 20% test (~572), preventing any look-ahead bias. StandardScaler (fit on training set only, applied to all splits) normalised all features to zero mean and unit variance, critical for the distance-sensitive SVM, KNN, and Logistic Regression models. Random Forest, being scale-invariant, was trained on unscaled features.

Feature Group	Features	Count
Trend (Moving Averages)	SMA_7, SMA_21, EMA_7, EMA_21	4
Momentum	RSI(14), MACD, MACD_Signal	3
Volatility (Bollinger)	BB_Upper, BB_Middle, BB_Lower	3
Price/Volume Returns	Daily_Return, Volume_Change, High_Low_Range, Open_Close_Diff	4
Lagged Returns	Return_Lag_1, _2, _3, _5	4

Table 1: Feature groups and composition (18 total features)

3. Feature Correlation Analysis

The Pearson correlation heatmap (Figure 3) reveals a near-perfect correlation cluster ($r \approx 1.0$) among SMA_7, SMA_21, EMA_7, EMA_21, and all three Bollinger Band levels. These features are redundant from an information-theoretic standpoint — all trace the level of the underlying price trend over similar windows — though their inclusion is justified by their widespread use in practitioner technical analysis. MACD and MACD_Signal share a very high mutual correlation ($r = 0.95$), while RSI correlates moderately with MACD ($r = 0.59$), reflecting shared momentum characteristics. Return-based and lagged features exhibit near-zero pairwise correlations, consistent with the weak-form EMH's assertion of serial return independence. This structure confirms that the feature set contains two largely orthogonal information channels: price-level trend indicators and return-momentum indicators.

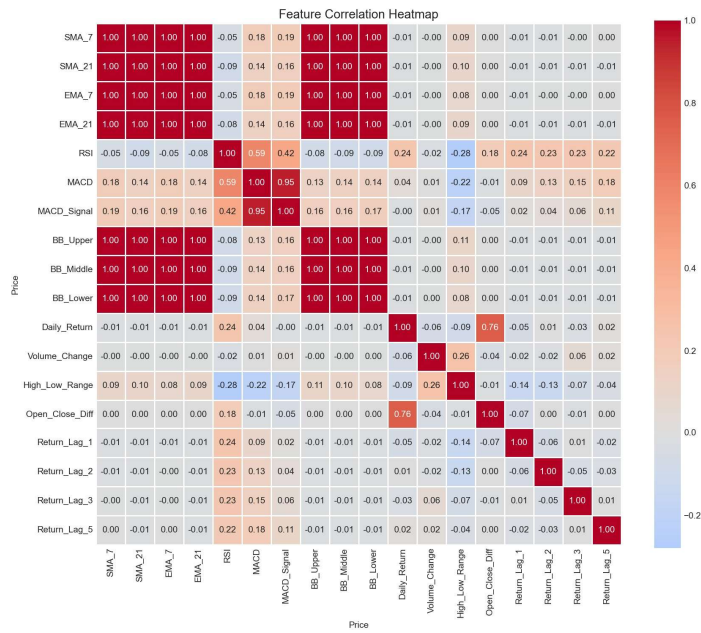


Figure 3: Pearson Correlation Heatmap of All 18 Features

4. Model Construction

Four classical supervised classifiers were selected to span a broad range of model complexity and learning philosophy. The Support Vector Machine (kernel='rbf', C=1.0, gamma='scale') constructs an optimal separating hyperplane in an implicit high-dimensional reproducing kernel Hilbert space, enabling non-linear decision boundaries with strong generalisation guarantees via margin maximisation — well-suited to low-signal financial data. The Random Forest (n_estimators=200, max_depth=10) builds an ensemble of deep decision trees on bootstrap-sampled data with random feature subsets, reducing variance through aggregation and yielding interpretable feature importances. K-Nearest Neighbours (k=5) is a non-parametric distance-based classifier that makes no distributional assumptions but is sensitive to feature scale and the curse of dimensionality in 18-dimensional space. Logistic Regression (max_iter=1000, L2 regularisation) serves as the linear baseline, estimating the probability of the UP class via a sigmoid of a weighted feature sum — its performance relative to non-linear models directly tests whether the predictive signal is linearly separable.

5. Evaluation and Results

Model	Val Accuracy	Test Accuracy	Rank
Support Vector Machine (SVM)	50.6%	52.1%	1st — Best
K-Nearest Neighbours (KNN)	49.0%	50.2%	2nd
Logistic Regression	50.3%	49.8%	3rd
Random Forest	49.4%	47.7%	4th

Table 2: Model Performance — Validation and Test Accuracy

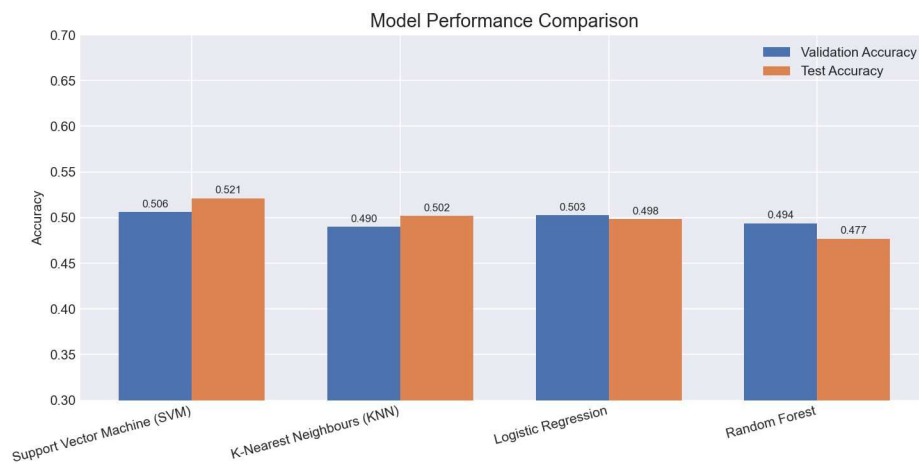


Figure 4: Model Performance Comparison — Validation vs. Test Accuracy

All four models cluster within $\pm 2.5\%$ of the 50% chance baseline, consistent with a largely efficient market for a highly liquid large-cap equity. The SVM achieved the highest test accuracy at 52.1%, benefiting from its non-linear RBF kernel and robust margin-based generalisation. KNN achieved 50.2%, while Logistic Regression and Random Forest performed below chance at 49.8% and 47.7% respectively. The below-chance Random Forest performance reflects mild overfitting: despite the `max_depth=10` constraint, 200 trees with 18 features appear to memorise noise patterns that do not generalise to the most recent market regime.

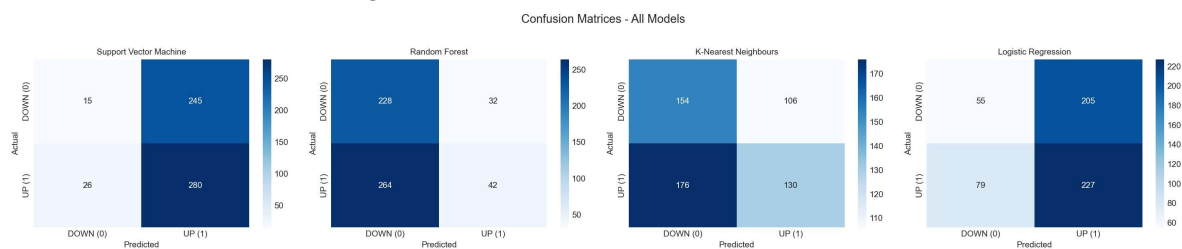


Figure 5: Confusion Matrices for All Four Models on Test Set

The confusion matrices (Figure 5) reveal a systematic UP-prediction bias across all models. The SVM, for instance, predicts UP for 525 of 566 test samples, correctly classifying 280 true UP days while mislabelling 245 actual DOWN days. This asymmetry reflects the net positive return tendency of AAPL over the evaluation window — models are largely learning the marginal class distribution (stock tends to go up) rather than genuinely discriminating between UP and DOWN conditions based on feature state. This "trivial predictor" effect is a well-documented phenomenon in financial classification literature and underscores the need for precision-recall and risk-adjusted metrics beyond raw accuracy.

5.1 Feature Importance (Random Forest)

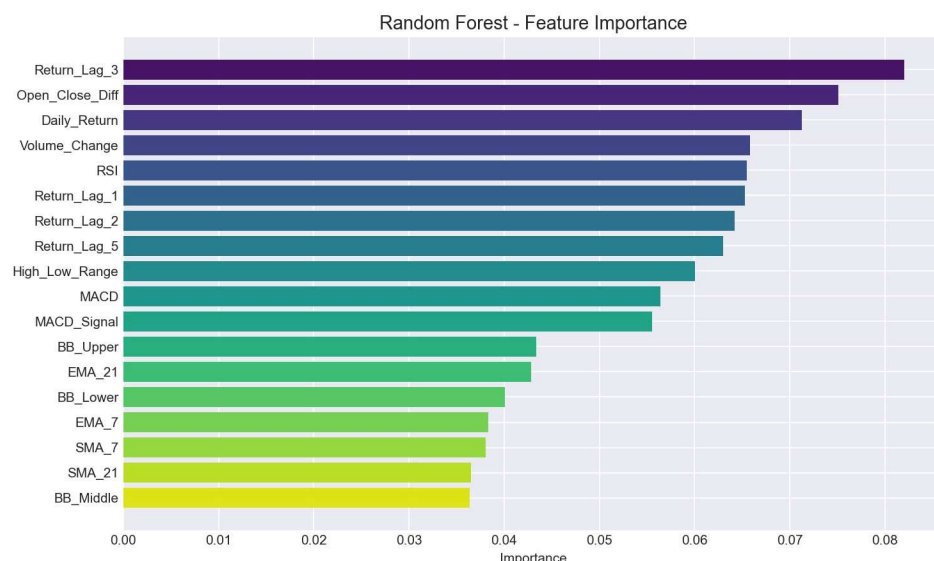


Figure 6: Random Forest Feature Importances — Mean Gini Impurity Decrease

The Random Forest importance scores (Figure 6) identify Return_Lag_3 (~8.2%), Open_Close_Diff (~7.4%), Daily_Return (~7.1%), Volume_Change (~6.6%), and RSI (~6.5%) as the five most discriminative features, collectively accounting for ~35.8% of total importance. The prominence of lagged returns confirms that short-term return autocorrelation — even if weak — represents the primary exploitable signal in this feature set. The trend-following features (SMA, EMA, Bollinger Bands) rank at the bottom (~3.5–4.3% each), reflecting their high multicollinearity: the Random Forest distributes their shared importance across redundant features, depressing individual scores. This analysis motivates a future feature selection step to eliminate redundant trend features and concentrate model capacity on the higher-information return and momentum signals.

6. Conclusion and Future Scope

This project has delivered a complete, reproducible end-to-end machine learning pipeline for next-day directional price prediction on AAPL, spanning data acquisition, feature engineering, temporal splitting, model training, and multi-model evaluation. The principal finding is clear: classical ML models operating on standard technical indicators achieve only marginal predictive accuracy above the 50% chance level (best: SVM at 52.1%), insufficient for economically significant trading signals once transaction costs are accounted for. This result is consistent with the semi-strong EMH for a heavily traded large-cap equity. Feature importance analysis reveals that return-momentum signals (lagged returns, RSI, daily return) carry more predictive power than trend-following indicators (moving averages, Bollinger Bands), which are largely redundant. The clustering of all model accuracies near 50% further suggests that model architecture is secondary to feature quality in this low-signal domain.

Future work should explore: (i) alternative data sources — earnings sentiment, options implied volatility, macroeconomic calendar events — to introduce orthogonal information channels; (ii) deep learning architectures (LSTM, TCN, Transformer) for end-to-end temporal pattern learning; (iii) walk-forward cross-validation for multi-regime evaluation; (iv) probabilistic position sizing using model confidence scores to improve risk-adjusted returns; and (v) dimensionality reduction (PCA) to eliminate the multicollinear trend-feature cluster and focus model capacity on genuinely independent signals.

7. References and Tools

Library / Tool	Version	Role
Python	3.10+	Primary language
yfinance	0.2.x	AAPL OHLCV data acquisition (Yahoo Finance API)
pandas / NumPy	2.x / 1.25+	Data manipulation and numerical computation
scikit-learn	1.3+	ML models, StandardScaler, metrics
Matplotlib / Seaborn	3.7+ / 0.12+	Visualisation and statistical plots
Jupyter Notebook	Any	Interactive development environment

Table 3: Software stack

Key References: Fama, E.F. (1970). Efficient Capital Markets. *Journal of Finance*, 25(2), 383–417. | Breiman, L. (2001). Random Forests. *Machine Learning*, 45(1), 5–32. | Cortes & Vapnik (1995). Support-Vector Networks. *Machine Learning*, 20(3), 273–297. | Murphy, J.J. (1999). *Technical Analysis of the Financial Markets*. NYIF.